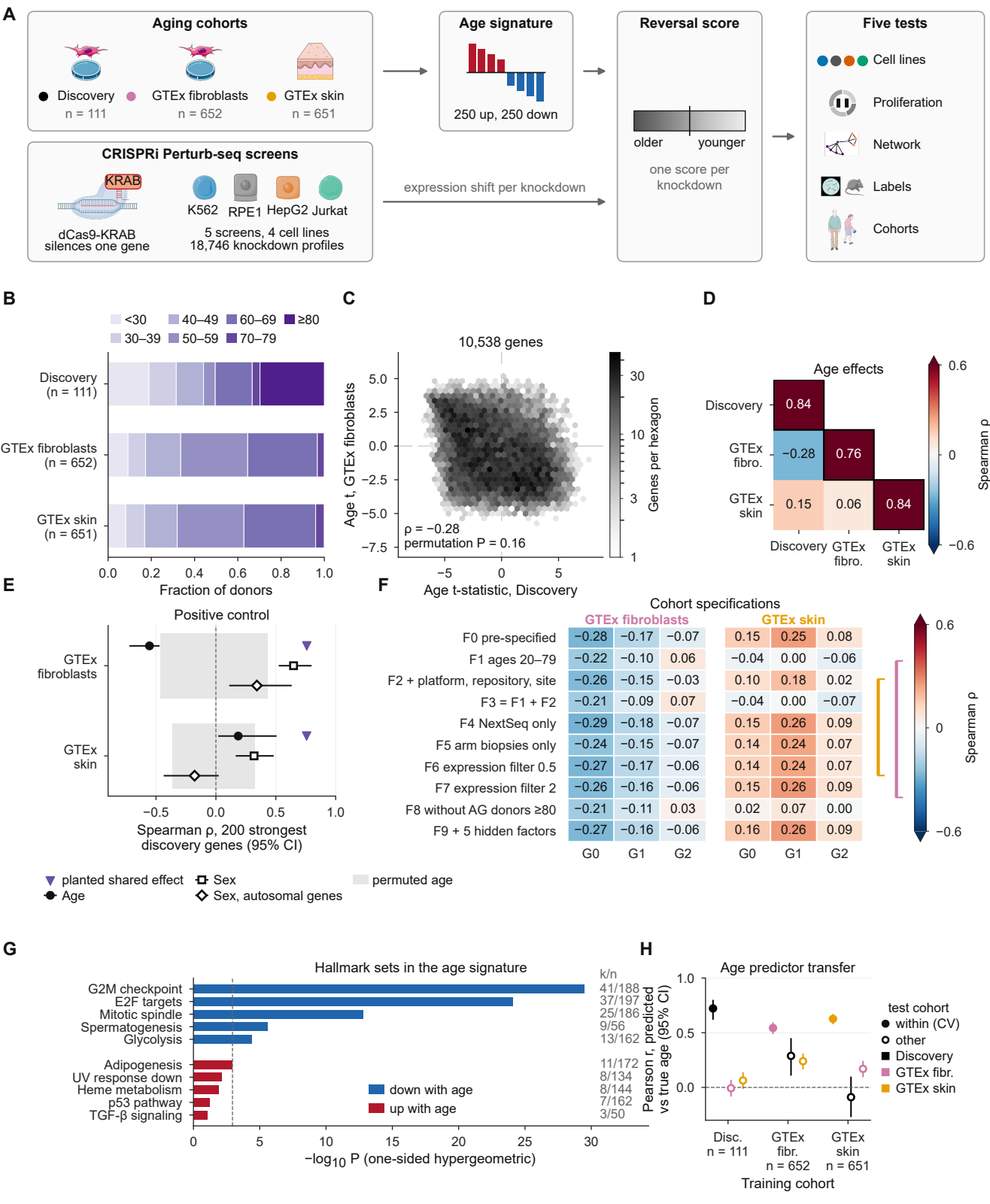

A
Aging cohorts
Discovery
n = 111
GTEx fibroblasts
n = 652
GTEx skin
n = 651
Age signature
250 up, 250 down
Reversal score
older
younger
one score per knockdown
Five tests
Cell lines
Proliferation
Network
Labels
Cohorts
CRISPRi Perturb-seq screens
KRAB
dCas9-KRAB silences one gene
K562
RPE1
HepG2
Jurkat
5 screens, 4 cell lines
18,746 knockdown profiles
expression shift per knockdown
B
<30
30–39
40–49
50–59
60–69
70–79
≥80
Discovery (n = 111)
GTEx fibroblasts (n = 652)
GTEx skin (n = 651)
Fraction of donors
C
10,538 genes
Age t, GTEx fibroblasts
Age t-statistic, Discovery
ρ = −0.28
permutation P = 0.16
Genes per hexagon
D
Age effects
Discovery
GTEx fibro.
GTEx skin
0.84
−0.28
0.76
0.15
0.06
0.84
Spearman ρ
E
Positive control
GTEx fibroblasts
GTEx skin
Spearman ρ, 200 strongest discovery genes (95% CI)
planted shared effect
Age
Sex
Sex, autosomal genes
permuted age
F
Cohort specifications
GTEx fibroblasts
GTEx skin
F0 pre-specified
F1 ages 20–79
F2 + platform, repository, site
F3 = F1 + F2
F4 NextSeq only
F5 arm biopsies only
F6 expression filter 0.5
F7 expression filter 2
F8 without AG donors ≥80
F9 + 5 hidden factors
G0
G1
G2
Spearman ρ
G
Hallmark sets in the age signature
G2M checkpoint
E2F targets
Mitotic spindle
Spermatogenesis
Glycolysis
Adipogenesis
UV response down
Heme metabolism
p53 pathway
TGF-β signaling
down with age
up with age
k/n
41/188
37/197
25/186
9/56
13/162
11/172
8/134
8/144
7/162
3/50
−log10 P (one-sided hypergeometric)
H
Age predictor transfer
Pearson r, predicted vs true age (95% CI)
Disc. n = 111
GTEx fibr. n = 652
GTEx skin n = 651
Training cohort
test cohort
within (CV)
other
Discovery
GTEx fibr.
GTEx skin